Problem 3

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Problem 1 *Given that:*

$\lim_{x\to-\infty} f(x) = 0$	$\lim_{x\to-3^-} f(x) = \infty$	$\lim_{x\to-3^+} f(x) = -\infty$	$\lim_{x\to4^-} f(x) = \infty$
$\lim_{x\to4^+} f(x) = -\infty$	$f(1) = 1$	$f(5) = -2$	$\lim_{x\to9} f(x) = 3$
$f'(1) \neq 0$	$f'(7) = 0$	$f'(x) = 2$ *for* $x > 9$	$f''(1) = 0$

the domain of f *is:* $(-\infty, -3) \cup (-3, 4) \cup (4, 9) \cup (9, \infty)$ *and* f *is continuous on its domain. The following sign chart for the first and second derivatives of* f*:*

find the following:

(a) *Critical points.*

(b) *Intervals where* f *is increasing/decreasing.*

(c) *Local extrema.*

(d) *Inflection points.*

(e) *Intervals of concavity.*

(f) *Sketch the graph of* f*.*

Explanation. (a) Critical points.

The critical points of f occur at points in the domain of f where either $f'(x) = 0$ or where $f'(x)$ does not exist. Even though $f'(-3)$, $f'(4)$, and $f'(9)$ do not exist, all three of those points are not in the domain of f and therefore are not critical points. We are given that $f'(7) = 0$, and so $x = 7$ is a critical point of f. Therefore, $x = 7$ is the only critical point of f.

(b) Intervals where f is increasing/decreasing.
f is increasing when $f'(x) > 0$. From the sign chart and our critical points, these are the intervals $(-\infty, -3)$, $(-3, 4)$, $(4, 7)$, and $(9, \infty)$. $f(x)$ is decreasing when $f'(x) < 0$. From the sign chart, this is on the interval $(7, 9)$.

(c) Local extrema.
Using the first derivative test, $f(7)$ is a local maximum because the derivative changes sign from positive to negative. So, $x = 7$ is the only local extremum of f, and it is a local maximum.

(d) Inflection points.
Possible inflection points occur where $f''(x) = 0$ or where $f''(x)$ does not exist. We are given that $f''(1) = 0$. In addition, $f''(x)$ does not exist at $x = -3, 4, 9$. However, these are not inflection points because f is not defined at these points. Since f'' changes sign at $x = 1$, this is in fact an inflection point of f. We are given that $f(1) = 1$, and so the only inflection point of f is the point $(1, 1)$.

(e) Intervals of concavity.
$f(x)$ is concave up when $f''(x) > 0$. From the sign chart, this is on the intervals $(-\infty, -3)$ and $(1, 4)$. $f(x)$ is concave down when $f''(x) < 0$. From the sign chart, this is on the interval $(-3, 1)$ and $(4, 9)$.

(f) Sketch the graph of f.